

Frederick Khasanto

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Education

The Hong Kong University of Science and Technology, Guangzhou *Sept 2026 – Jun 2030*

Ph.D. in Intelligent Transportation

- Intelligent Logistics and Supply Chain Analysis (iLSCA) Lab supervised by Prof. Benedict Jun MA
- Postgraduate Studentship Recipient

The Chinese University of Hong Kong, Shenzhen *Sept 2022 – Jun 2026*

B.Sc. in Data Science and Big Data Technology

- CGPA: 3.74 / 4.00 — MGPA: 3.74 / 4.00 (First Class Honours)
- **Relevant Courses:** Mathematical Statistics, Optimization, Advanced Machine Learning, Reinforcement Learning, Stochastic Processes & Simulation, Natural Language Processing, Deep Learning and Applications

Research and Projects

Research Experience

Research Assistant – Offensive Speech Detection Project, CUHK(SZ) *Sep 2025 – Nov 2025*

- Conducted literature review on transformer-based hate-speech detection under Prof. Benyou Wang research group.
- Collected existing datasets for supervised classification.

Summer Research Intern – VLAA Lab, University of California, Santa Cruz *Jul 2025 – Aug 2025*

- Reviewed vision-language model literature (e.g., LLaVA, LLaVa-CoT) and multimodal architectures.
- Summarized findings under supervision of Hardy Chen and Prof. Cihang Xie.

Selected Course Projects

Diffusion Models for X-ray Image Generation - Deep Learning Project *Oct 2025 - Dec 2025*

- Implemented DDPM for medical X-ray image generation in PyTorch.

KnowYourBody – AI Practicum Project *Oct 2025 - Dec 2025*

- Developed a web app integrating LLM APIs for educational and personalized health analytics.

SmartBot: Cloud-based Clinic Appointment Chatbot - Cloud Computing Project *Oct 2024 – Dec 2024*

- Developed an appointment scheduling chatbot using AWS Cloud Functions.

Teaching

Undergraduate Student Teaching Fellow - DDA4002: Stochastic Simulation *Jan 2026 – May 2026*

- Led tutorials on input and output analysis, simulation optimization, and stochastic gradient descent.
- Conducted a midterm exam review session covering key topics and practice problems.

Undergraduate Student Teaching Fellow - STA2002: Probability and Statistics II *Sep 2025 – Dec 2025*

- Led tutorials on point estimation, interval estimation, hypothesis testing, and order statistics.
- Conducted a midterm exam review session covering key topics and practice problems.

Undergraduate Student Teaching Fellow - STA2002: Probability and Statistics II *Jan 2025 – May 2025*

- Led weekly tutorials on statistical methods including Chi-square tests, ANOVA, linear regression, and likelihood ratio tests.
- Conducted a final exam review session covering key topics and practice problems.
- Designed two homework assignments on hypothesis testing, ANOVA, and regression analysis.

Undergraduate Student Teaching Fellow -*Jan 2024 – May 2024***DDA3010: Applied Machine Learning**

- Held office hours to assist students in understanding course materials.

Undergraduate Student Teaching Fellow -*Sep 2023 – Dec 2023***CSC1001: Introduction to Programming Methodology**

- Led weekly Python tutorials for first-year students, covering control flow, data structures, and functions.
- Created problem sets and live-coding examples to support learning and engagement.
- Graded assignments and provided feedback to help students improve coding and debugging skills.

Leadership and Service

Peer Advisor Group Leader*Sep 2024 – Jul 2025**School of Data Science, CUHK(SZ)*

- Coordinated group meetings between academic advisors, peer mentors, and freshmen to foster communication and academic guidance.
- Mentored first-year students on academic planning, study strategies, and campus adaptation.

Vice Chair, Academic and Career Support Dept.*Aug 2023 – Jul 2024**International Students' Association, CUHK(SZ)*

- Organized mentoring programs, academic talks, and alumni panels to support students' academic and career development.
- Coordinated company sharing sessions and site visits to provide exposure to real-world industry practices.

University Library Student Assistant*Feb 2023 – May 2024**CUHK(SZ)*

- Assisted with material cataloging and provided logistical support for library workshops and student events.

Awards and Scholarships

- Outstanding Graduate Award - Muse College (2026)
- Dean's List, School of Data Science (2023, 2024, 2025)
- Excellent Student Leader Award - University Honors (2025)
- Outstanding Leadership Award – Muse College (2024, 2025)
- Shenzhen Universiade International Student Scholarship (2024)
- Academic Performance Scholarship Class C (2023)
- 3rd Prize – CUHK(SZ) Programming Contest (2023)
- Guangdong Government Outstanding International Student Scholarship (2022, 2023)
- CUHK(SZ) Full Tuition and Accommodation Scholarship (2022–2026)

Technical Skills

Programming: Python, R, C++, SQL**Frameworks:** PyTorch, Scikit-learn, Pandas, NumPy**Tools:** Git, LaTeX**Languages:** English (Fluent), Chinese (Working Proficiency)